

A TRAINING, once per building, on one fault-free period

1 Trend data + configuration file

Temperatures, flows, pressures, damper and valve positions, fan speeds, usually once a minute.

The configuration file maps sensor names to roles and lists the control sequence as rules. It is the only building-specific part.

2 Events along the control sequence

Each day is one case.
State events: what the equipment is doing (“cooling started at 09:14”).

Signature events: something that should not happen (“flow through a closed valve”). Never used for model discovery.

3 Process models of a normal day

Discovered from the state events only, so the models cannot learn our own fault rules.

Device level: one model per terminal unit.
Unit level: one model for the air handler’s whole day.

4a Calibrate the eight checks

The last eight days of each month (96 in a year) are held out and used only to measure each check’s false-alarm rate, the rate reported with every alarm.

Healthy bands and model thresholds are set on the other days.

B DEPLOYMENT, every day

(2) New day of data

Turned into events with the same configuration.

4b The eight checks, each against its own healthy band

Read the raw sensors

Rules: a control-sequence rule broken
Residual: a physical relation drifted
Oscillation: actuator reversing too often

Read the event log

Unit conformance: day fits unit model
Device conformance: day fits device model
Absence: expected device silent
Frequency: a daily count out of range

Rate (support only): a mode’s 30-day activity against the same calendar month of the fault-free year

5 Noise gate

A check may alarm only when its flags exceed what its own false-alarm rate would produce by chance; the rate is never taken below 3 days in 96.

6 Alarm to the operator

What broke
Where (unit, terminal unit, zone)
Evidence (values and duration)
The check’s measured false-alarm rate

models and thresholds from A

false-alarm rates from A